ARCHI2
Homework Assignment

Learning Objectives:

- Design basic interrupt systems to meet elementary control requirements
- Design an elementary sequencer meeting given specifications

WARNING: Any plagiarism will result in an automatic grade of zero and a penalty will be applied to the global final assignment.

Due Date: Thursday 8 January 2026 at 23:59

Part A: Hierarchical Interrupt System Design

Problem Statement

You are tasked with designing a calculator system to manage a company. The system must have a hierarchical interrupt system capable of handling the following:

- **Power failures** (Level 0 – Highest Priority): Alert the system of a power cut to allow necessary backups using emergency batteries
- **Disk management** (Level 1): The system must manage 4 priority disks required for calculator operation
- **Alarm system** (Level 2): The calculator will also manage an alarm system to monitor up to 6 rooms. Initially, only 4 rooms are connected, with 2 additional rooms pending wiring
- **Reserved level** (Level 3 – Lowest Priority): A final level is available to connect up to 4 new interrupt sources if necessary

Questions:

1. **(30 points)** Propose a complete schematic for the interrupt system to be implemented. You must:
   - Clearly identify all interrupt sources and connect them appropriately
   - Specify the content of important registers at system startup
2. **(20 points)** Propose an extension to your schematic to allow direct hardware recognition of the highest priority interrupt level. Do not use a predefined logisim circuit to implement the extension, rather use a minimum elementary circuits/gates.

Part B: Daisy-Chain Recognition Circuit

Problem Statement

For recognizing interrupt causes of the Input/Output level (Level 1), a Daisy-Chain circuit is used.

Each Daisy-Chain cell is connected to an interrupt signal generated by one of the Input/Output controllers. The Daisy-Chain cell operates as follows:
When an interrupt signal is received, the CPU sends an acknowledgment signal **equal to "1"**. When a cell receives the acknowledgment signal, three scenarios arise:

1. If the controller, that is connected to the cell, has sent an interrupt signal **AND** the input acknowledgment signal equals **"1"**, then it is the highest priority cell that sent an IT signal. It places the number of the cause that is associated to the controller that is connected to it on the Data Bus and sets the output acknowledgment signal to **"0"**

2. If the controller, that is connected to the cell, has sent an interrupt signal **BUT** the input acknowledgment signal equals **"0"**, then a higher priority cell has sent an interrupt signal. Nothing happens, and the output acknowledgment signal remains at **"0"**

3. If the controller, that is connected to the cell, has not sent an interrupt signal, then it passes the input acknowledgment signal to the output unchanged (if it is **"1"**, it remains **"1"**; if it is **"0"**, it remains **"0"**)

Questions:

1. **(25 points)** Design the detailed internal schematic of the Daisy-Chain system. Your design should include:

   - All input signals (4 interrupt requests, input acknowledgment)
   - All output signals (interrupt request output, output acknowledgment, 2-bit device number)
   - Internal logic implementing the three operational scenarios

2. **(15 points)** Connect the Daisy-Chain system to the circuit of Part A, to inform the sequencer of the highest priority interrupt cause number. Show how the different cells are cascaded and connected to:

   - The system data bus
   - The CPU acknowledgment signal
   - Each other in a daisy-chain configuration

Part C: Sequencer Implementation

Problem Statement

For the Input/Output interrupt level (Level 1), the interrupt service routine address is stored in a ROM.
The interrupt source number deposited by the Daisy-Chain cell on the data bus is used by the sequencer as an address in the ROM. The retrieved address is then loaded into the Program Counter register.

The machine's Data-Path is shown in the following diagram:

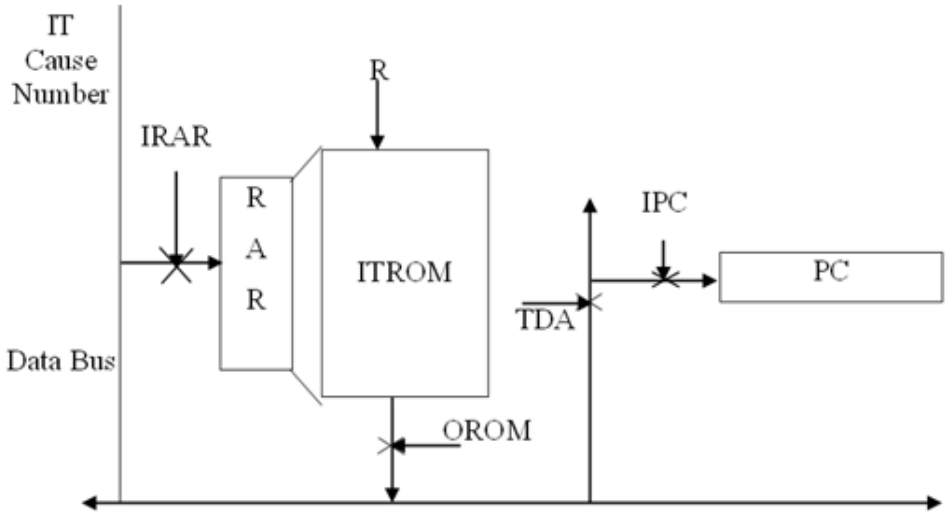


Figure 1: Machine's Data-Path

The ROM configuration containing the interrupt service routine addresses is as follows:

Cause Number	Memory Address
0	00001111
1	00011100
2	00110000
3	01000110

Questions:

1. **(30 points)** Design a 4-phase distributor generating:

   - 4 non-overlapping phases: Φ_0, Φ_1, Φ_2, Φ_3
   - 4 pulses: Θ_0, Θ_1, Θ_2, Θ_3

   Requirements:

   - Each phase should be active for one clock cycle
   - Phases should follow sequentially: $\Phi_0 \rightarrow \Phi_1 \rightarrow \Phi_2 \rightarrow \Phi_3 \rightarrow \Phi_0$
   - Each pulse Θ_i should be a short pulse occurring during phase Φ_i

2. **(30 points)** Implement the ROM circuit using transistors.

3. **(50 points)** Implement the sequencer circuit that performs a Jump to the interrupt service routine based on the interrupt number generated by the Daisy-Chain circuit, using the phase distributor designed previously.

4. Draw a timing diagram manually (without using the Logisim facilities), that shows the sequence of the different stages as according to the clock signals. Add the diagram to the project to upload.

5. Connect all system components into a single Logisim circuit.

   Requirements: Your implementation should:

   - Show the control signals required for each step
   - Implement the complete interrupt handling sequence:
     - Step 0: Recognize interrupt level
     - Step 1: Recognize interrupt cause number from Daisy-Chain
     - Step 2: Access ROM to get service routine address
     - Step 3: Load new address into PC
   - Provide a clear schematic showing all connections
   - Implement the complete design in Logisim.

Submission Requirements

- All schematics must be clearly labeled with signal names.
- Test the functionality of the circuits by simulating the occurrence of an Interrupt signal.
- You must upload the Logisim circuits on the Classroom platform. A score will be assigned.
- The filename of your project must follow this format: "***LastName_ FirstName_ GroupNumber.circ***".
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